C 21835 (Pages : 3) Name.....

Reg. No.....

SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2022

Physics/Applied Physics

PHY 2B 02/APY 2B 02—PROPERTIES OF MATTER WAVES AND ACOUSTICS

(2016—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions*
1 *marks each.*

1. S.I. unit of strain is ————————.
2. Relation between Y, K and σ is ————.
3. What is the equation of bending moment of a beam ?
4. Time for one complete oscillation is called ————.
5. For a wave of period 10 seconds, frequency = ————— Hz
6. What is damping constant ?
7. In longitudinal wave motion, Particles of the ———— vibrate ———— the direction of wave.
8. Define stationary wave.
9. A given periodic function is capable of Fourier analysisv if the function is ———— and ————.
10. Decibel is the unit of ————.

(10 × 1 = 10 marks)

Section B

Answer **all** *questions.*
Write in two **or** *three sentenses.*
2 *marks each.*

11. What is a torsion pendulum ?
12. Derive the differential equation of motion for a simple harmonic oscillator.

Turn over

13. Derive the theoretical limiting values of Poisson's ratio.
14. Explain principle of superposition.
15. Show that in a damped harmonic oscillator velocity decreases exponentially with time.
16. What is angle of twist ?
17. What is anharmonic oscillator ?

(7 × 2 = 14 marks)

Section C

Write any **five** *questions.*
Write in one paragraph.
4 *marks each.*

18. Write and explain the equation of differential equation of damped harmonic oscillator.
19. Differentiate between wave velocity and group velocity.
20. Obtain the expression for velocity of longitudinal waves in rods.
21. State Fourier theorem. Explain its importance and limitations.
22. Derive the relation between elastic constants and Poisons ratio.
23. Differentiate between reflection at rigid boundary and free boundary.
24. Derive the equation for work done work done in a torsion pendulum

(5 × 4 = 20 marks)

Section D

Solve any **four** *problems.*
4 *marks each.*

25. CalculateThePoisson's ratio for steel, given that its Youngs modulus is $2\times 10^{11}Nm^{-2}$ and rigidity modulus is $8 \times 10^{10}Nm^{-2}$.
26. Calculate the work done in twisting a steel wire of diameter 2 mm. and length 0.25 m. through an angle 45°. Given rigidity modulus for steel is $8 \times 10^{10}Nm^{2}$.
27. A steel tap is stretched so that its length is 0.001 % of its initial line. If Y = 200 GPa, Calculate the stress required

28. A plane progressive wave traveling along positive X direction has amplitude 0.2 cm, velocity 360 cm/s and wavelength 60 cm. (a) Write down its equation when (1) the displacement is zero at $x = 0$ and $t = 0$; (2) displacement is maximum at $x = 0$ and $t = 0$. (b) Also find the displacement in either case at $x = 120$ cm and $t = 2$ *s*.

29. Calculate the tensile stress in a wire of density 8000 kgm^{-3}.When the velocity of transverse wave on it is Qqual to 340 ms^{-1}.

30. Find a Fourier's series for the function defined by the equations

    Y=

    1 for t=0 to t = T/2

    0 for t = T/3

    + 1 for t = T/2 to t = T

31. Prove that the velocity of a particle in S.H.M. at a distance $\sqrt{3}/2$ of its amplitude from the centre is half of its velocity at the central position.

(4 × 4 = 16 marks)

Section E

Write any **two** *questions.*

10 *marks each.*

32. Explain the composition of two rectangular simple harmonic motions of equal periods Define forced harmonic oscillator.

33. What is damping force ? Derive the equation for damped harmonic oscillator. Explain over damped and under damped cases.

34. Study the variation of KE, PE and Total energy of SHM with displacement.

35. Explain in detail,acoustics of buildings.

(2 × 10 = 20 marks)